Digital Forensics CYBI 4319

Lab 02: Evidence Collection & MBR Analysis

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Scenario/Case:

Last February, there was a possible hacking attempt in a school district. After tracking the suspect for multiple months, the authorities were able to find his residence although they were unsuccessful at finding the criminal, they found a USB drive in his living room next to his desk. Digital specialists were tasked to find his content.

Executive summary

In this report about a possible extension case, we analyzed evidence related to MBR and GPT. The collected evidence was checked for integrity, and chain of custody by confirming cryptographic hashes of the evidence before and after compression. Analysis of the evidence was performed using tools like Linux, FTK imager, and HxD. This report details our findings in the USB.

Introduction

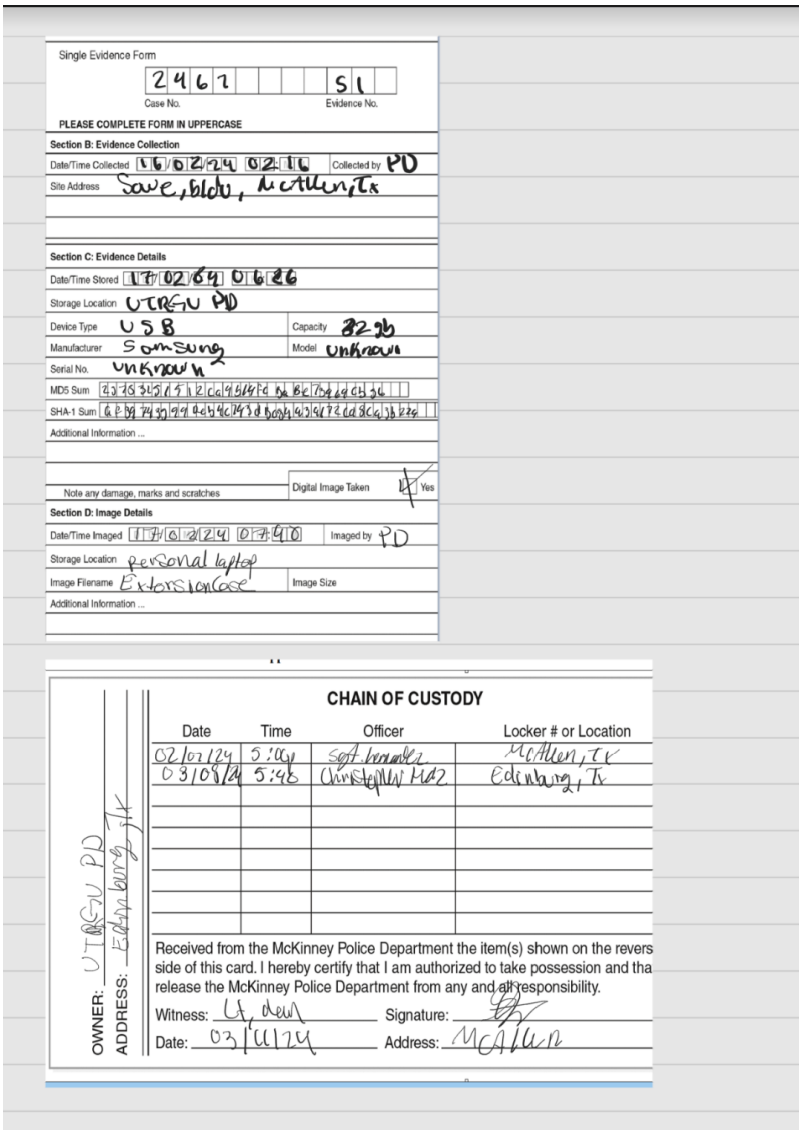
After a possible attempt, I analyzed the contents of a USB drive and I

Evidence collection:

digital first responder collects evidence by assessing the scene, documenting it, ensuring safety, isolating affected devices, and using forensically sound methods to collect digital evidence. This includes creating forensic images, using write-blocking devices, and collecting volatile data if applicable. They maintain a chain of custody for all evidence, document the collection process, package the evidence securely, and hand it over to appropriate personnel for further analysis and investigation. Adherence to established guidelines and best practices is crucial to ensure the integrity and admissibility of the evidence in legal proceedings.

Chain of custody:

Due to the intervention of federal agencies, the local police department got the device almost a year after the attach took place, here. The digital investigator, christopher hernandez got it last for examination.



Single Evidence Form

Case No. 2462 Evidence No. S1

PLEASE COMPLETE FORM IN UPPERCASE

Section B: Evidence Collection

Date/Time Collected 16/02/24 02:16 Collected by PD

Site Address Save, blvd, McAllen, Tx

Section C: Evidence Details

Date/Time Stored 17/02/24 06:26

Storage Location UTRGV PD

Device Type USB Capacity 32gb

Manufacturer Samsung Model Unknown

Serial No. Unknown

MD5 Sum [illegible]

SHA-1 Sum [illegible]

Additional Information ...

Note any damage, marks and scratches Digital Image Taken ☒ Yes

Section D: Image Details

Date/Time Imaged 17/02/24 07:40 Imaged by PD

Storage Location personal laptop

Image Filename Extorsioncase Image Size

Additional Information ...

CHAIN OF CUSTODY

Date	Time	Officer	Locker # or Location
02/02/24	5:00	Sgt. hernandez	McAllen, TX
03/08/24	5:48	Christopher MDZ	Edinburg, Tx

Received from the McKinney Police Department the item(s) shown on the revers side of this card. I hereby certify that I am authorized to take possession and tha release the McKinney Police Department from any and all responsibility.

Witness: Lt. dew Signature:

Date: 03/11/24 Address: McAllen

OWNER: UTRGV PD

ADDRESS: Edinburg, TX

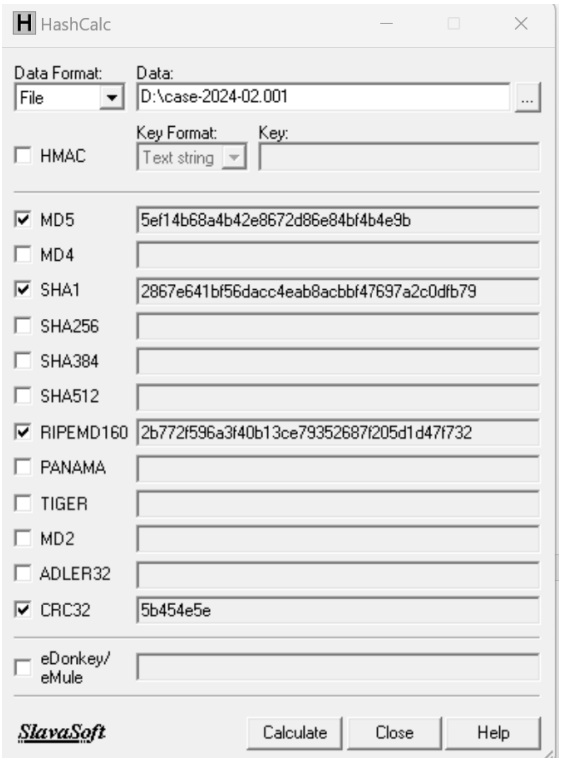
MD5 (Message Digest Algorithm 5) and SHA (Secure Hash Algorithm) are cryptographic hash functions used to produce a fixed-size hash value from input data of any size. MD5 produces a 128-bit hash value and is commonly used for checksums and data integrity verification. However, MD5 is considered cryptographically broken and unsuitable for further use due to vulnerabilities. SHA has several versions, with SHA-1, SHA-256, SHA-384, and SHA-512 being commonly used. SHA-256, for example, produces a

256-bit hash value and is widely used for secure data transmission, digital signatures, and password hashing due to its stronger security properties compared to MD5.

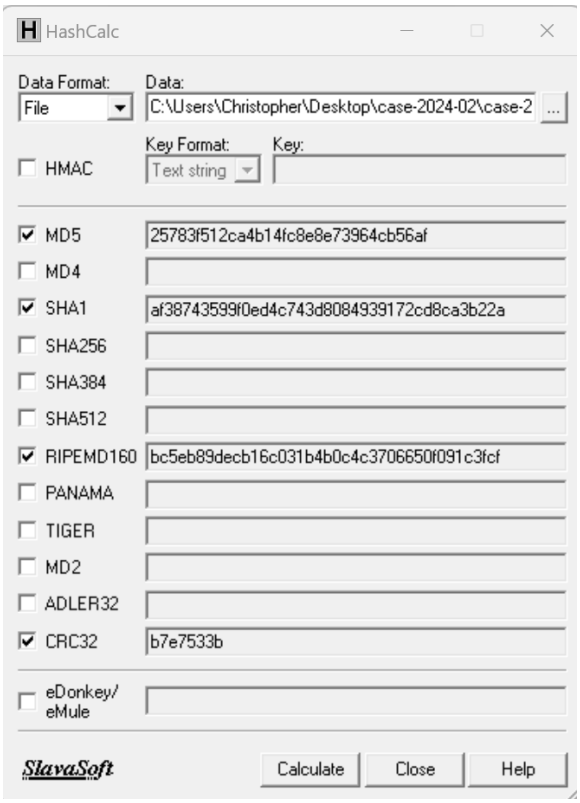
Hases:

Hashes are fixed-length strings of characters generated from input data using a hash function. They are used in various applications for data integrity verification, password hashing, digital signatures, and more. Hash functions take an input (or message) of any size and produce a fixed-size output (the hash value or hash digest), typically a sequence of hexadecimal digits. Hash functions have several important properties, including determinism (the same input always produces the same hash), pre-image resistance (it is computationally infeasible to determine the input from the hash), and collision resistance (it is computationally infeasible to find two different inputs that produce the same hash). Hashes are widely used in computer science and cryptography for data security and integrity.

This is the MD5 and SHA of the file after unzipping it:



This is before:



Evidence analysis:

Here are the partitions and what they mean:

Partition 1:

80 01 01 00 07 0B 02 89 3F 00 00 00 C1 97 21 00

Boot Indication (Not Bootable or Bootable): The first byte 80 indicates that this partition is bootable.

Head - Starting Number: The second byte 01 represents the starting head of the partition.

Sector - Starting Number: The third byte 01 represents the starting sector of the partition.

Cylinder - Starting Number: The fourth byte 00 and the fifth byte 07 together represent the starting cylinder of the partition.

System Indicator (Partition Type): The sixth byte 0B represents the partition type. In this case, 0B typically indicates a FAT32 file system.

Head - Ending Number: The seventh byte 02 represents the ending head of the partition.

Sector - Ending Number: The eighth byte 89 represents the ending sector of the partition.

Cylinder - Ending Number: The ninth byte 3F and the tenth byte 00 together represent the ending cylinder of the partition.

Number of Sectors Before Partition (Relative Sectors): The eleventh byte 00 00 00 represents the number of sectors before the partition starts. In this case, it's 0.

Number of Sectors in Partition (Length): The twelfth byte C1 97 21 represents the length of the partition in sectors. When converted to decimal, this is approximately 342001 sectors.

Size of the Partition in Bytes: To calculate the size of the partition in bytes, you would multiply the number of sectors (342001 in this case) by the sector size (typically 512 bytes). The size would be approximately 342001 * 512 = 175,398,912 bytes.

The drive's total capacity can be determined by looking at the entire disk's size. However, this information is not provided in the partition entry alone. To find the total capacity of the drive, you would need to examine the disk's partition table, looking at the sizes of all partitions and possibly the total size of the disk itself.

Partition 2:

00 0B 03 89 07 F4 63 11 00 98 21 00 00 90 21 00

Boot Indication: Not bootable (00)

Start: Head=0B, Sector=03, Cylinder=89

System Indicator: Unknown (F4)

End: Head=63, Sector=11, Cylinder=00

Sectors Before Partition: 3,224,961 sectors

Sectors in Partition: 9,812,224 sectors

Size of the Partition: 9,812,224 sectors

Partition 3:

00 F4 64 11 OC FE 7F EA 00 28 43 00 00 40 35 00

Boot Indication: Not bootable (00)

Head - Starting Number: F4

Sector - Starting Number: 64

Cylinder - Starting Number: 11

System Indicator (Partition Type): FAT32 (0C)

Head - Ending Number: FE

Sector - Ending Number: 7F

Cylinder - Ending Number: EA

Number of Sectors Before Partition (Relative Sectors): 28,675,328 sectors

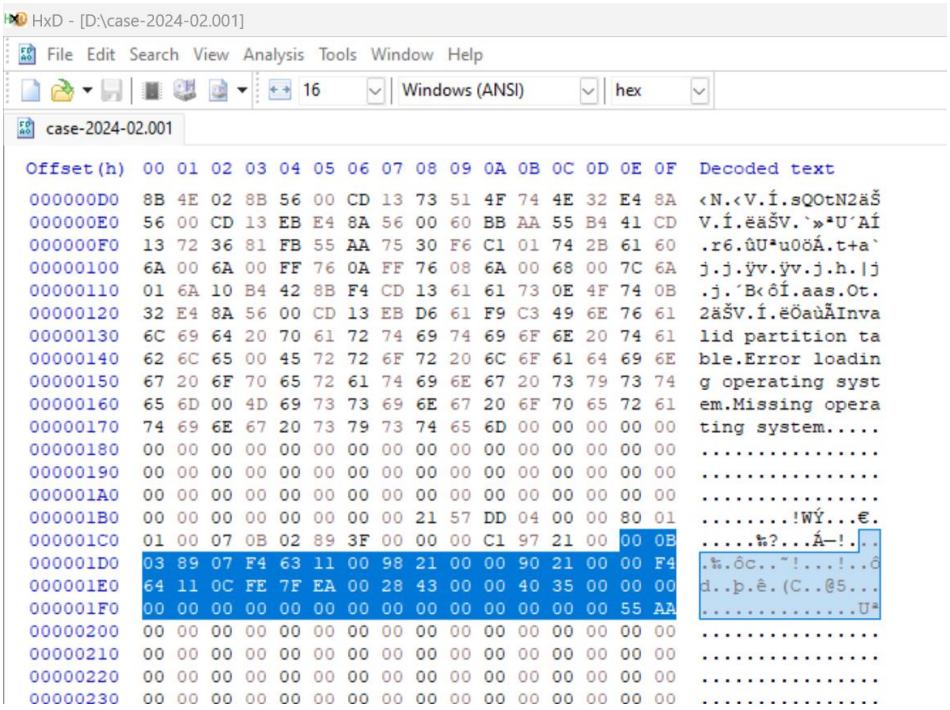
Number of Sectors in Partition (Length): 3,325,056 sectors

Size of the Partition in Bytes: (3,325,056 sectors

Partition 4:

This partition is all zeros, this doesnt not add any value to the investigation as it is normal.

HxD analysis:



To validate the hashes, I used

I used Linux to verify the disk signature.

```
┌──(kali㉿kali)-[~/digitalforensics/linuxsidefolder2]
└─$ sudo fdisk -l
Disk /dev/sda: 80.09 GiB, 86000000000 bytes, 167968750 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1d2d6b62

Device     Boot Start       End   Sectors  Size Id Type
/dev/sda1  *     2048 167968749 167966702 80.1G 83 Linux
```

Overall analysis:

I found that

Conclusion:

Due to the hack of illegally trying to access students' information, trying to hack a school district, I have examined a USB memory and analyzed the contents of it. I have found that the USB memory does not hold any important information to identify the suspect nor know where he is at.

References

Awati, R. (2022, May 19). What is The master boot record (MBR)?. WhatIs. https://www.techtarget.com/whatis/definition/Master-Boot-Record-MBR

Wikimedia Foundation. (2024, February 27). Master boot record. Wikipedia. https://en.wikipedia.org/wiki/Master_boot_record#References